



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

YEARLY EXAMINATION

PRELIMINARY ASSESSMENT

TASK 4

Year 11

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 – 14
- Write your Board of Studies Student Number **and** Class Teacher on the writing booklet(s) **or** sheet(s) submitted
- **Weighting:** 40%

Total marks – 70

Section I

Pages 3 – 7

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 8 – 14

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

Section I 10 marks

- Circle the correct response on the answer sheet provided
 - Each question is worth 1 mark
-

1. For what values of k does the equation $x^2 - 6x - 3k = 0$ have real roots?

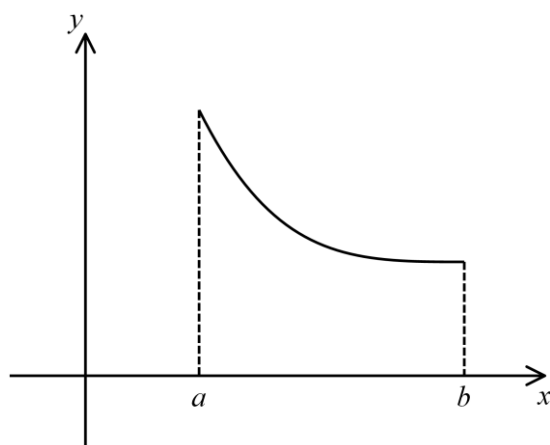
(A) $k \geq -3$

(B) $k \leq -3$

(C) $k \geq 3$

(D) $k \leq 3$

2. For the function $y = f(x)$, $a < x < b$ graphed below:



which of the following is true?

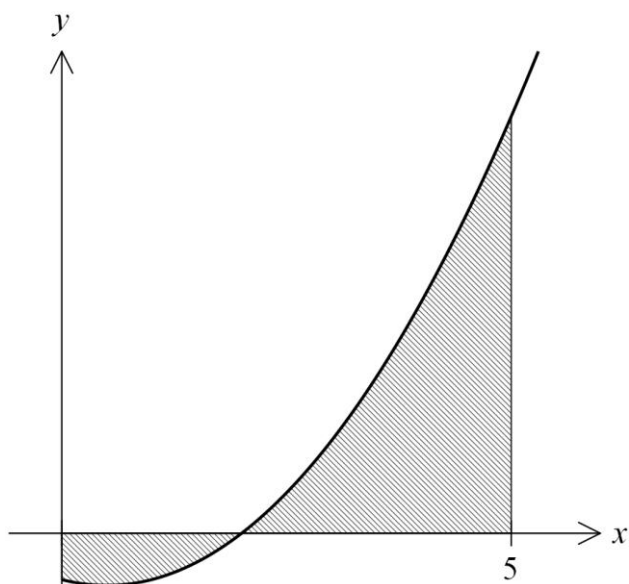
(A) $f'(x) > 0$ and $f''(x) > 0$

(B) $f'(x) > 0$ and $f''(x) < 0$

(C) $f'(x) < 0$ and $f''(x) > 0$

(D) $f'(x) < 0$ and $f''(x) < 0$

3. Which expression will give the area of the shaded region bounded by the curve $y = x^2 - x - 2$, the x -axis and the lines $x = 0$ and $x = 5$?



- (A) $A = \left| \int_0^1 (x^2 - x - 2) dx \right| + \int_1^5 (x^2 - x - 2) dx$
- (B) $A = \int_0^1 (x^2 - x - 2) dx + \left| \int_1^5 (x^2 - x - 2) dx \right|$
- (C) $A = \left| \int_0^2 (x^2 - x - 2) dx \right| + \int_2^5 (x^2 - x - 2) dx$
- (D) $A = \int_0^2 (x^2 - x - 2) dx + \left| \int_2^5 (x^2 - x - 2) dx \right|$
4. The solution to the equation $e^x = 4$, to three significant figures, is:
- (A) 1.386
- (B) 1.38
- (C) 1.39
- (D) 54.598

5. $\sum_{n=3}^7 2n + 3 =$

(A) 17

(B) 65

(C) 77

(D) 91

6. What are the coordinates of the focus of the parabola $4y = x^2 - 8$?

(A) (0, -8)

(B) (0, -7)

(C) (0, -2)

(D) (0, -1)

7. What are the domain and range of the function $f(x) = \sqrt{4 - x^2}$?

(A) Domain: $-2 \leq x \leq 2$; Range: $0 \leq y \leq 2$

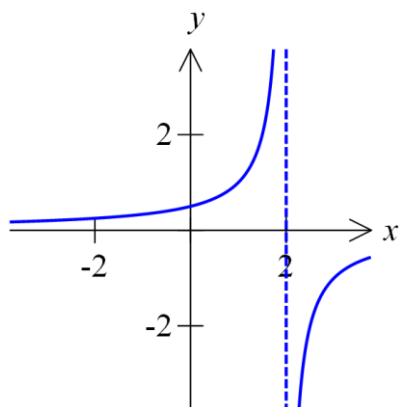
(B) Domain: $-2 \leq x \leq 2$; Range: $-2 \leq y \leq 2$

(C) Domain: $0 \leq x \leq 2$; Range: $-4 \leq y \leq 4$

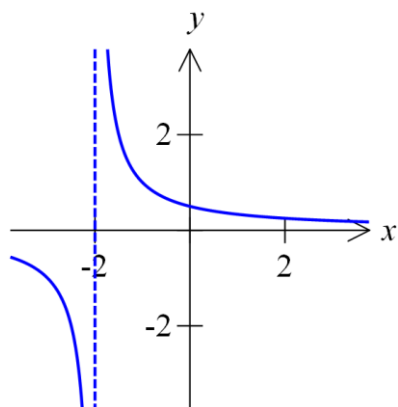
(D) Domain: $0 \leq x \leq 2$; Range: $0 \leq y \leq 4$

8. Which of the following represents the graph of $y = \frac{1}{2-x}$?

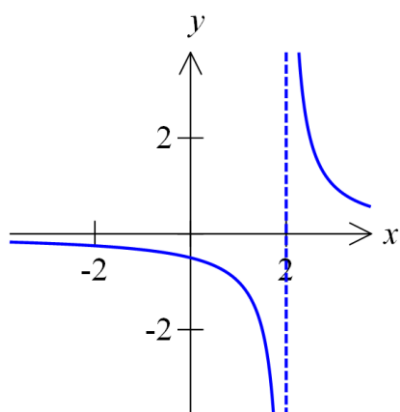
(A)



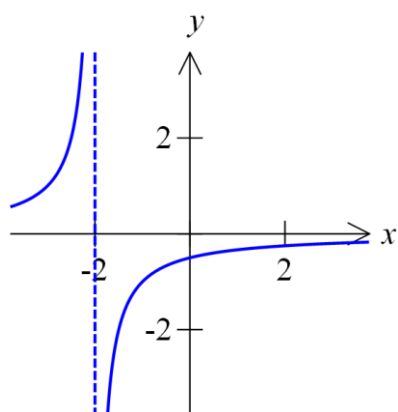
(B)



(C)



(D)



9. What is the value of $\int_0^1 (e^{3x} - 1) dx$?

(A) $\frac{e^3}{3}$

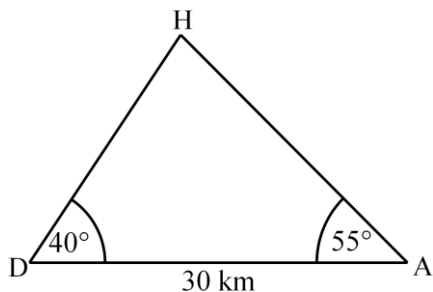
(B) $\frac{e^3}{3} - 1$

(C) $e^3 - 1$

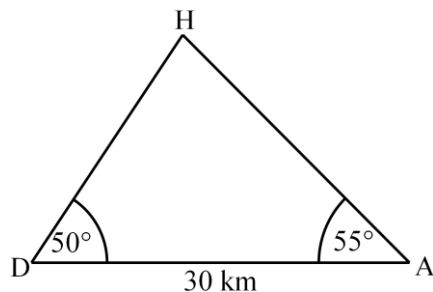
(D) $\frac{1}{3}(e^3 - 4)$

10. Archer (A) is 30km due east of Dulwich (D). Holwood (H) is on a bearing of 040° from Dulwich and 325° from Archer. Which of the following diagrams best represents this information?

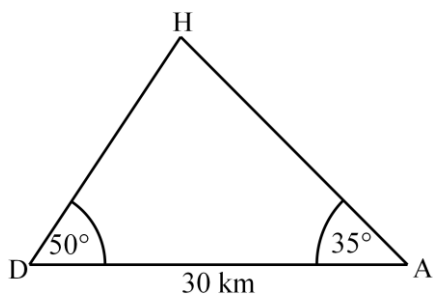
(A)



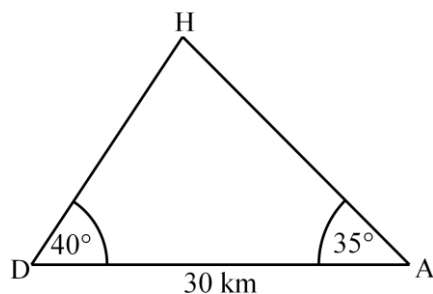
(B)



(C)



(D)



End of Section I

Section II 60 marks

- Begin each question in a new writing booklet or on a new answer sheet
 - Show all necessary working
 - Each question is worth 15 marks
-

Question 11 (15 marks)

a) Differentiate

i) $\sqrt{2x - x^2}$ 2

ii) $e^{4x} + \frac{1}{x}$ 2

iii) $\frac{3x - 7}{3 + 4x}$ 2

b) Find:

i) $\int (5x - 1)^9 dx$ 2

ii) the value of k such that $\int_1^k \frac{4}{x^2} dx = 3$. 2

c) Solve $2\sin x + \sqrt{3} = 0$ for $0^\circ \leq x \leq 360^\circ$. 3

d) Find the centre and radius of the circle $x^2 - 6x + y^2 + 2y - 6 = 0$. 2

Question 12 (15 marks) Begin a NEW answer booklet or answer sheet

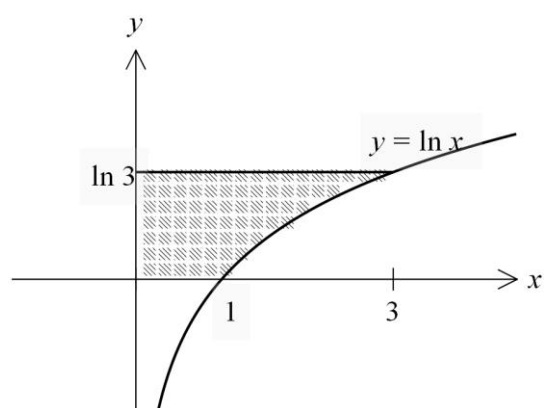
- a) The sum of the first n terms of an arithmetic series is given by:
- $$S_n = n(38 - 2n)$$
- i) Calculate S_1 and S_2 . **1**
- ii) Find the first three terms of the series. **2**
- iii) Find an expression for the n -th term. **2**
- b) A parabola has the equation $2y^2 = 4x + 8$.
- i) Express this equation in the form $(y - y_1)^2 = 4a(x - x_1)$. **2**
- ii) Draw a neat sketch of the parabola showing the focus; vertex; axis of symmetry and x and y -intercepts. **2**
- c) Use Simpson's Rule with 5 function values to find an approximate value of **3**

$$\int_2^4 \frac{x}{2} \ln x \, dx$$

correct to 3 decimal places.

Question 12 continues on the following page.

- d) The diagram shows the area bounded by the graph $y = \ln x$, the coordinate axes and line $y = \ln 3$.

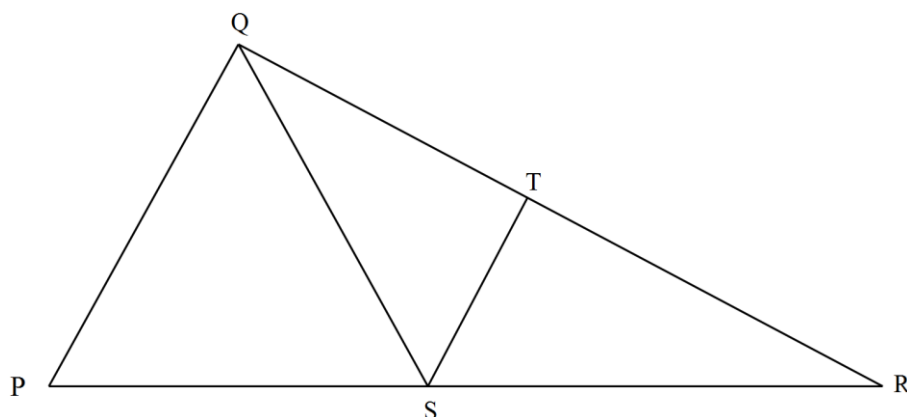


- i) Find the shaded area. **2**

- ii) Hence write down the exact value of $\int_1^3 \ln x \, dx$. **1**

Question 13 (15 marks) Begin a NEW answer booklet or answer sheet

- a) In $\triangle PQR$, point T lies on side QR and point S lies on side PR such that $QT = TR$, $QS = QP$ and $ST \perp QT$.



Copy or trace the diagram in your answer booklet.

- | | | |
|---|---|---|
| i) | Prove that $\triangle QTS \equiv \triangle RTS$. | 2 |
| ii) | Prove that $\angle QPS = 2\angle TQS$. | 2 |
| b) Consider the function $f(x) = x^4 - 4x^3$. | | |
| i) | Find the stationary points on the curve and determine their nature. | 4 |
| ii) | Find any points of inflexion. | 2 |
| iii) | Sketch the curve $f(x) = x^4 - 4x^3$ showing all important features, including x and y -intercepts and stationary points. | 2 |
| c) A normal is drawn to the curve $y = e^{2x}$ at the point $P(\log_e 2, 4)$. The normal cuts the x -axis at Q . | | |
| i) | The equation of the normal at P is given by $x + 8y = k$ where k is a real number. Find the exact value of k . | 2 |
| ii) | Find the coordinates of Q . | 1 |

Question 14 **(15 marks)** **Begin a NEW answer booklet or answer sheet**

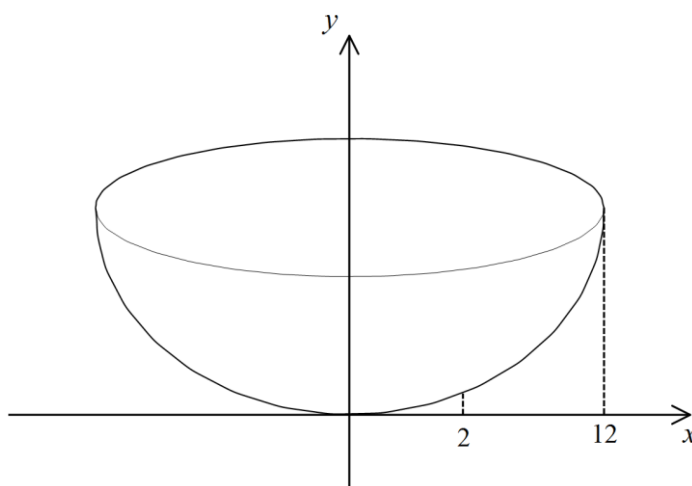
a) Given the equation $3x^2 + 4x - 3 = 0$ has roots α and β , evaluate the following:

i) $\alpha + \beta$ and $\alpha\beta$ **1**

ii) $(\alpha - 1)(\beta - 1)$ **1**

iii) $\alpha^3 + \beta^3$ **1**

b) A bowl was designed by rotating the section of the curve $y = \frac{1}{4}x^2$ between $x = 2$ cm and $x = 12$ cm, about the y -axis.



i) Calculate the volume of the bowl, leaving your answer in terms of π . **2**

ii) Hence, calculate the capacity of the bowl, correct to the nearest litre. **1**
(1000 cm³ = 1 litre)

Question 14 continues on the following page.

- c) Penny has just commenced work and is investigating superannuation funds. She calculates that she will need \$1 000 000 in her fund when she retires in 40 years. She finds a fund guaranteeing to pay interest on each deposit at the rate of 6% p.a. compounded monthly.

She intends to deposit \$ M into the fund at the beginning of each month for 40 years.

After 25 years she intends to double the amount she deposits each month. That is, she will deposit a further \$ M into the fund for the remaining 15 years.

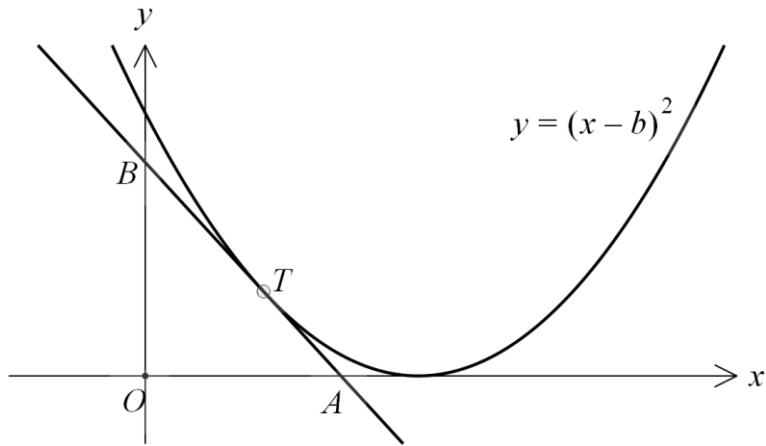
- i) Show that the value of M can be found using the formula **2**

$$1000000 = 201M(1.005^{480} + 1.005^{180} - 2).$$

- ii) Find the value of M . **1**

Question 14 continues on the following page.

- d) The diagram shows the graph of $y = (x - b)^2$.
 $T(t, (t - b)^2)$ is a point on the curve such that $0 < t < b$.
 The tangent at T is drawn to intersect with the x and y axes at the points A and B respectively as shown.



- i) Show that the tangent to the curve at T is given by $y = (2x - t - b)(t - b)$. 2
- ii) Find the coordinates of the point T , in terms of b , which will **maximise** the area of $\triangle ABO$. 4

End of Section II

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(\sqrt{x^2 - a^2} + x \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(\sqrt{x^2 + a^2} + x \right)$$

Note $\ln x = \log_e x, \quad x > 0$



TRINITY GRAMMAR SCHOOL
2012, Year 11 Mathematics Extension 1
Yearly Examination
SECTION I
ANSWER SHEET

Student Number: **Class Teacher:**

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

-
- | | | | | |
|-------------|---|---|---|---|
| Q1. | A | B | C | D |
| Q2. | A | B | C | D |
| Q3. | A | B | C | D |
| Q4. | A | B | C | D |
| Q5. | A | B | C | D |
| Q6. | A | B | C | D |
| Q7. | A | B | C | D |
| Q8. | A | B | C | D |
| Q9. | A | B | C | D |
| Q10. | A | B | C | D |

YEAR 11 EXTENSION 1 ASSESSMENT 2 (1/3/12)

Question 1

a) $f(x) = x^2 + 1 + x^{-2}$
 $f(1+\sqrt{2}) = (1+\sqrt{2})^2 + 1 + (1+\sqrt{2})^{-2}$ ✓
 $= 1 + 2\sqrt{2} + 2 + 1 + \frac{1}{1+2\sqrt{2}+2}$ ✓
 $= 4 + 2\sqrt{2} + \frac{1}{3+2\sqrt{2}} \times \frac{(3-2\sqrt{2})}{(3-2\sqrt{2})}$ ✓
 $= 4 + 2\sqrt{2} + \frac{3-2\sqrt{2}}{9-8}$
 $= 7.$

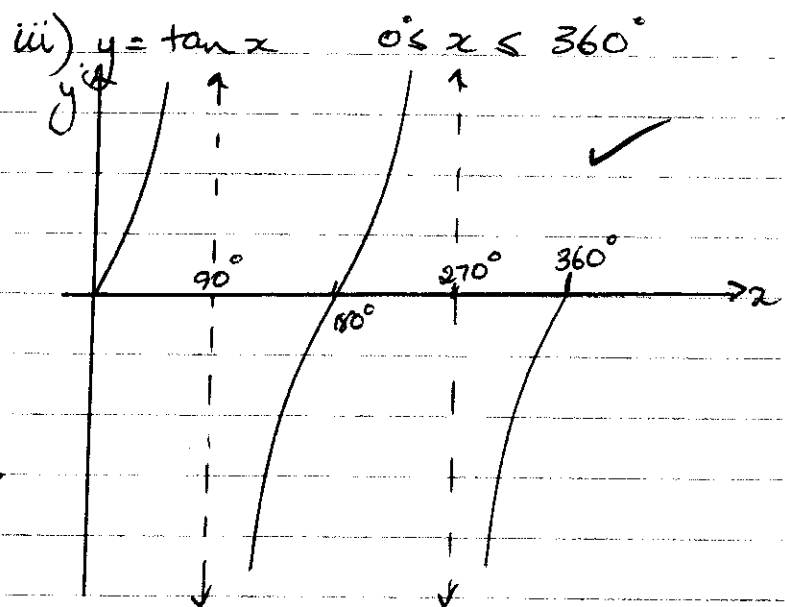
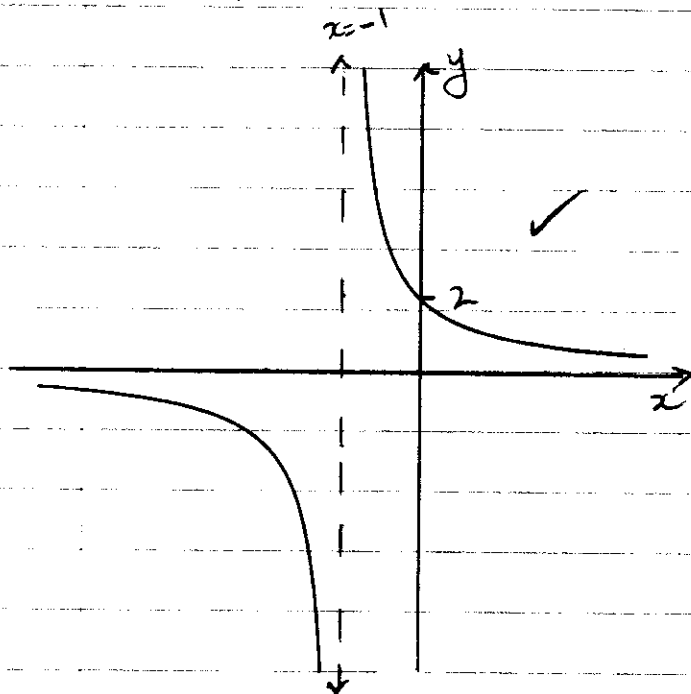
b) $x^2 - 10x + y^2 + 6y + 33 = 0$
 $x^2 - 10x + 25 + y^2 + 6y + 9 = -33 + 25 + 9$
 $(x-5)^2 + (y+3)^2 = 1$
 $\therefore$ centre $(5, -3)$ ✓ $r = 1$ ✓

c) $y = \frac{x}{4-x^2}$ D: $4-x^2 \neq 0$ ✓
 $x^2 \neq 4$
 $x \neq \pm 2$ ✓
 $\therefore$ D: all real x , except $x \neq \pm 2$.

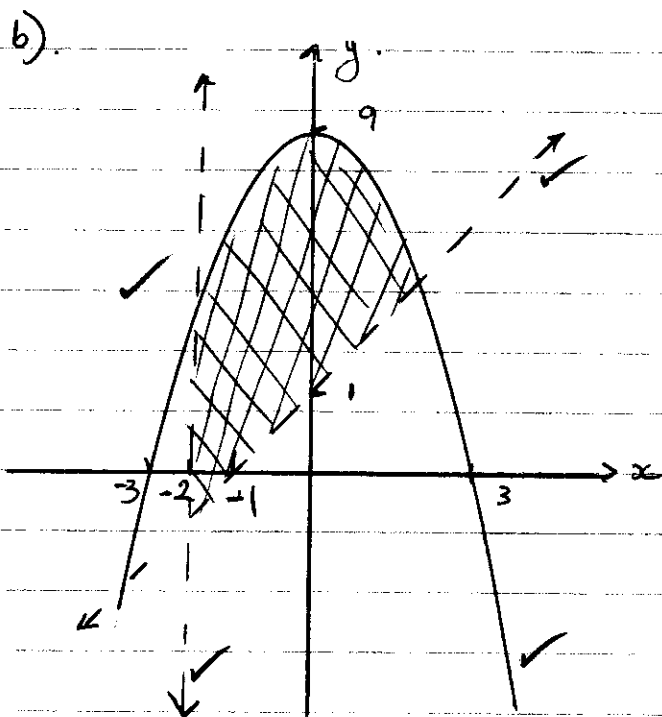
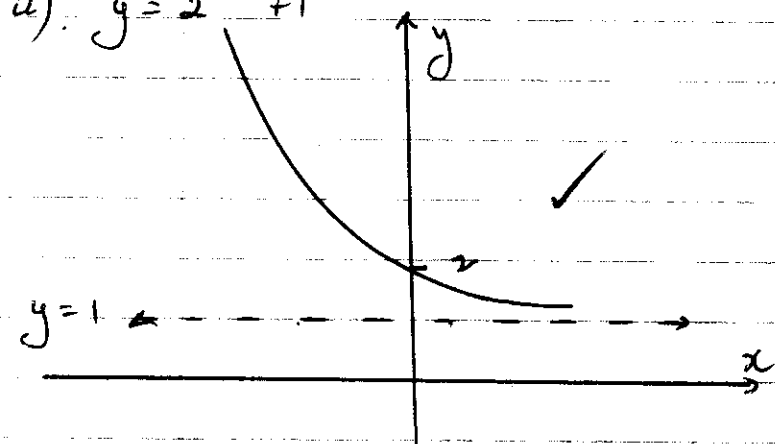
d) $f(x) = x - x^2$
 $f(-x) = (-x) - (-x)^2$
 $= -x - x^2$
 $-f(x) = -(x - x^2)$
 $= -x + x^2$
 Since $f(x) \neq f(-x)$
 or is $-f(x) \neq f(-x)$
 the function is neither even or odd. ✓

Q2

a) i) $y = \frac{2}{x+1}$



ii) $y = 2^{-x} + 1$



$$\text{Q2 c) i)} f(1) = b(1)^2 + 2 \cdot a$$

$$= b + 2$$

$$f(-1) = (-1 - 1)^2$$

$$= 4$$

$$\therefore b + 2 = 4$$

$$\underline{b = 2}$$

$$f\left(\frac{1}{2}\right) = a\left(\frac{1}{2}\right) + 3$$

$$f\left(-\frac{1}{2}\right) = \left(-\frac{1}{2} - 1\right)^2$$

$$= \frac{9}{4}$$

$$\therefore \frac{a}{2} + 3 = \frac{9}{4}$$

$$2a + 12 = 9$$

$$2a = -3$$

$$a = -\frac{3}{2}$$

$$\text{ii)} \quad f(x) = \begin{cases} (x-1)^2 & x < 0 \\ -\frac{3}{2}x + 3 & 0 \leq x < 1 \\ 2x^2 + 2 & x \geq 1 \end{cases}$$

$$\therefore \text{Range: } y > 1$$

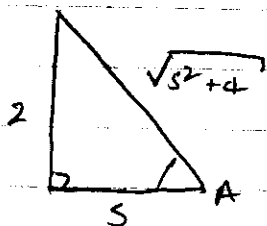
Q3

$$\begin{aligned}
 \text{a) } \cos 225^\circ \times \tan 150^\circ \\
 &= \cos(180+45) \times \tan(180-30) \\
 &= -\cos 45^\circ \times -\tan 30^\circ \\
 &= -\frac{1}{\sqrt{2}} \cdot -\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{6}} = \frac{\sqrt{6}}{6} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \tan \theta &= -1.44598 \\
 \theta &= 180 - 55.33, 360 - 55.33 \\
 &= 124^\circ, 304^\circ \checkmark
 \end{aligned}$$

$$\text{c) } \sec A = \frac{\sqrt{s^2+4}}{s}$$

$$\therefore \cos A = \frac{s}{\sqrt{s^2+4}}$$



$$\cos^2 A = \frac{s^2}{s^2+4} \checkmark$$

$$\sin^2 A = \left(\frac{2}{\sqrt{s^2+4}} \right)^2 = \frac{4}{s^2+4}$$

$$\begin{aligned}
 \sin^2 A + \cos^2 A &= \frac{4}{s^2+4} + \frac{s^2}{s^2+4} \\
 &= \frac{s^2+4}{s^2+4} = 1 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{d) } \cot^2 \theta &= \frac{1}{\sin^2 \theta} \\
 &= \frac{\cos^2 \theta}{\sin^2 \theta} - \frac{1}{\sin^2 \theta} \\
 &= \frac{\cos^2 \theta - 1}{\sin^2 \theta} \checkmark
 \end{aligned}$$

$$= \frac{\sin^2 \theta}{\sin^2 \theta} = 1 \checkmark$$

$$\text{e) } \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$$

$$\text{LHS} = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$= \frac{1 - \tan^2 \theta}{\sec^2 \theta} \checkmark$$

$$= (1 - \tan^2 \theta) \div \frac{1}{\cos^2 \theta}$$

$$= \left(1 - \frac{\sin^2 \theta}{\cos^2 \theta} \right) \cdot \cos^2 \theta \checkmark$$

$$= \cos^2 \theta - \frac{\sin^2 \theta \cos^2 \theta}{\cos^2 \theta}$$

$$= \cos^2 \theta - \sin^2 \theta = \text{RHS}$$

Question 4.

a)

i) In ΔABC .

$$\frac{BC}{\sin 25} = \frac{20}{\sin 13} \quad \checkmark$$

$$BC = \frac{20 \sin 25}{\sin 13}$$

ii) $\sin 38 = \frac{CD}{BC}$

$$CD = BC \sin 38 \quad \checkmark$$
$$= \frac{20 \sin 38 \sin 25}{\sin 13}$$

$$= 23.13301 \dots$$

$$\approx 23 \text{ m.}$$

b) ΔADC : $A = \frac{1}{2} ab \sin C$.

$$= \frac{1}{2} \cdot 9 \cdot 8 \cdot \sin 38$$
$$= 22.16 \text{ cm}^2 \quad \checkmark$$

In ΔABC :

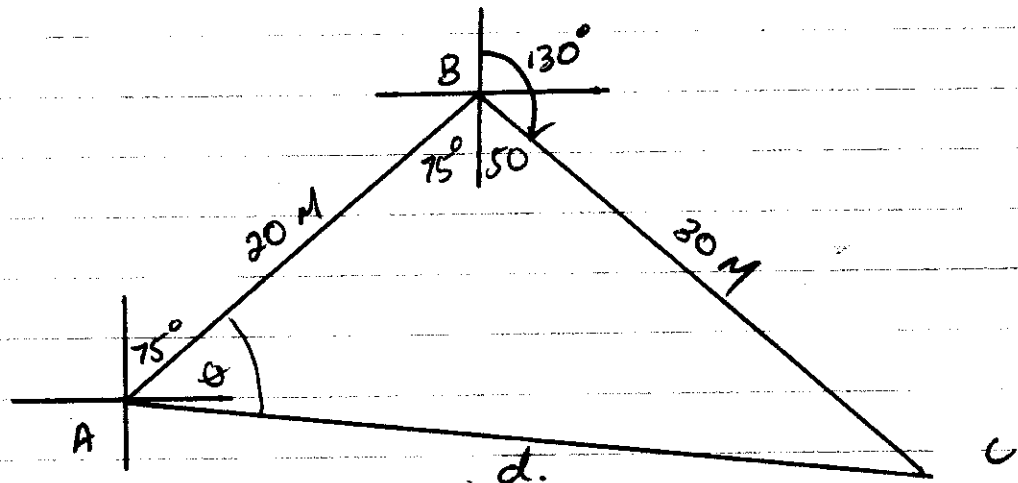
$$BC^2 = 6^2 - 4^2$$
$$= 20 \quad \checkmark$$

$$BC = \sqrt{20}$$

$$\text{Area} = \frac{1}{2} \cdot 4 \cdot \sqrt{20}$$
$$= 8.94 \text{ cm}^2 \quad \checkmark$$

$$\therefore \text{Total Area} = 22.16 + 8.94$$
$$= 31.10808 \dots$$
$$= 31.11 \text{ cm}^2 \quad \checkmark$$

c)



i)
$$d^2 = 20^2 + 30^2 - 2(20)(30)\cos(125) \quad \checkmark$$

$$= 1988.29$$

$$d = 44.590264 \dots$$

$$= 45 \text{ nautical miles.} \quad \checkmark$$

ii)
$$\frac{\sin \theta}{30} = \frac{\sin 125}{45}$$

$$\sin \theta = \frac{30 \sin 125}{45}$$

$$\theta = 33^\circ \quad \checkmark$$

Bearing = $33 + 75 = 108 \quad \checkmark$
 $\therefore$ Bearing from A to C. is 108° T.